UNIVERSITY EXAMINATION

COURSE CODE: CSC 301

COURSE TITLE: Structured Programming & Data Structures I

Time Allowed: 2 Hours
Total Marks: 40

Instructions: Answer ALL questions. Each question carries equal marks. Choose the most appropriate answer for each multiple-choice question.

1. Which of the following is NOT a fundamental control structure in structured programming?
    1. Sequence
    2. Selection
    3. Iteration
    4. Encapsulation

2. In C/C++, what does the

&

operator do when placed before a variable name?
    1. Dereferences a pointer
    2. Returns the memory address of the variable
    3. Performs a logical AND operation
    4. Declares a reference variable only

3. A data structure where elements are accessed in a Last-In-First-Out (LIFO) manner is called a:
    1. Queue
    2. Tree
    3. Stack
    4. Graph

4. What is the time complexity of accessing a specific element in an array using its index?
    1. O(1)
    2. O(n)
    3. O(log n)
    4. $O(n^2)$

5. Which data structure is most naturally suited for implementing a printer spooler or a task scheduling queue?

1. Stack
2. Queue
3. Linked List
4. Array

6. In a singly linked list, each individual node contains:

1. Data and two pointers
2. Only data
3. Data and a pointer to the next node
4. Data and a pointer to the previous node

7. Which of the following searching algorithms strictly requires the target array to be sorted beforehand?

1. Linear Search
2. Binary Search
3. Depth-First Search
4. Breadth-First Search

8. What is the worst-case time complexity of the Bubble Sort algorithm?

1. O(n)
2. O(n log n)
3. $O(n^2)$
4. O(log n)

9. The computational process of finding a specific target element within a data structure is known as:

1. Sorting
2. Searching
3. Traversing
4. Inserting

10. A function that directly or indirectly calls itself during execution is referred to as a:

1. Iterative function
2. Recursive function
3. Inline function
4. Virtual function

11. What is the primary purpose of a "base case" in a recursive function?

1. To define the maximum number of recursive calls
2. To act as the condition under which the recursion stops

3. To initialize the variables used in the recursion
4. To throw an exception if an error occurs

12. Which of the following is classified as a non-linear data structure?

1. Array
2. Stack
3. Queue
4. Tree

13. How do you correctly declare a pointer variable designed to hold the address of an integer in C?

1. int p;
2. int *p;
3. int &p;
4. pointer p;

14. What value does the

malloc()

function return if it fails to allocate the requested memory?

1. 0
2. -1
3. NULL
4. void

15. What is the main structural advantage of a linked list over a standard array?

1. It allows for O(1) random access to elements
2. It consumes significantly less memory overhead
3. It allows for dynamic resizing without contiguous memory allocation
4. It provides faster binary search capabilities

16. In a simple array-based implementation of a queue, what happens when the rear index reaches the end of the array but the front index is not at the beginning (0)?

1. A Queue Full error occurs, wasting empty space
2. The array is automatically resized by the operating system
3. Elements seamlessly wrap around to the beginning
4. The program crashes due to segmentation fault

17. Which standard operation is used to retrieve and remove an item from the top of a stack?

1. push()
2. pop()
3. enqueue()

4. dequeue()

18. What is the numerical index of the first element in an array in C-based programming languages?

1. 1
2. 0
3. -1
4. Variable depending on the compiler

19. Conceptually, a two-dimensional array can be most accurately thought of as a:

1. Linear list of elements
2. Tree of interconnected elements
3. Table consisting of rows and columns
4. Directed graph

20. Which of the following is NOT typically considered an advantage of structured programming paradigms?

1. Easier to read and understand code
2. Easier to systematically test and debug
3. Encourages heavy use of global variables
4. Promotes modularity and code reusability

21. What specific condition leads to a "stack overflow"?

1. The stack is entirely empty and a pop operation is attempted
2. The stack pointer points precisely to the bottom of the stack
3. More items are continuously pushed onto the stack than its allocated capacity can hold
4. Memory is freed automatically before it should be

22. The standard operation to append an element to the rear of a queue is known as:

1. Push
2. Pop
3. Enqueue
4. Dequeue

23. In a standard circular linked list, the "next" pointer of the final (last) node points to:

1. NULL
2. The first (head) node
3. The immediately preceding node
4. Itself

24. What is the fundamental difference between a

struct

and an array in C?

1. Structs can aggregate multiple disparate data types, whereas arrays aggregate elements of a single uniform data type
2. Arrays are always dynamically allocated, whereas structs are always static
3. Structs are reserved exclusively for logical operations, arrays for data storage
4. There is no functional difference; they are aliases

25. The algorithmic process of rearranging data items into a specified ordered sequence (e.g., ascending or descending) is called:

1. Searching
2. Traversing
3. Sorting
4. Merging

26. Which of the following sorting algorithms operates by continuously scanning the list and swapping adjacent elements if they are found to be in the wrong relative order?

1. Selection Sort
2. Insertion Sort
3. Bubble Sort
4. Merge Sort

27. A pointer variable that stores the memory address of another pointer variable is formally known as a:

1. Double pointer
2. Reference pointer
3. Array pointer
4. Void pointer

28. Which of the following symbolic operators represents a logical OR operation in the C language?

1. &&
2. &
3. ||
4. |

29. In the context of dynamic memory, what constitutes a "memory leak"?

1. When a program legally utilizes too much system RAM
2. When dynamically allocated memory is never properly freed or deallocated after its useful life
3. When a program crashes abruptly due to insufficient physical memory
4. When secure memory sectors are accessed illegally out of bounds

30. In C programming, which built-in function is utilized to explicitly release previously dynamically allocated memory back to the system?

1. delete()
2. remove()
3. free()
4. clear()

31. A selection (or decision) construct in structured programming is best represented by:

1. A for loop
2. A while loop
3. An if-else statement
4. Sequential arithmetic instructions

32. The specific term "dereferencing" a pointer describes:

1. Assigning a brand-new memory address to the pointer variable
2. Accessing or manipulating the actual value stored at the memory address the pointer holds
3. Freeing the memory block the pointer references
4. Declaring the pointer in the global scope

33. Which of the following well-known algorithms employs a "divide-and-conquer" algorithmic strategy?

1. Bubble Sort
2. Insertion Sort
3. Binary Search
4. Linear Search

34. When an interpreter or compiler evaluates a complex arithmetic expression (like converting infix to postfix), which data structure is most critically utilized?

1. Queue
2. Tree
3. Linked List
4. Stack

35. The size (number of elements) of a standard static array in C must be explicitly known at:

1. Runtime
2. Compile time
3. Execution time
4. Load time

36. Which keyword is specifically used to assign a new, convenient name to an existing data type or a user-defined struct in C?

1. datatype
2. type
3. typedef
4. structdef

37. A standard singly linked list can be sequentially traversed in how many directions?

1. One direction only (forward)
2. Two directions (forward and backward)
3. Three directions
4. Any arbitrary direction

38. What is the time complexity of appending a new element to the very end of a linked list if we actively maintain a dedicated pointer to the tail node?

1. O(1)
2. O(n)
3. O(log n)
4. $O(n^2)$

39. In a standard

while

loop structure, at what point is the conditional expression evaluated?

1. After the loop body fully executes
2. Before the loop body is permitted to execute
3. Halfway through the loop's execution cycle
4. Only if a runtime error is encountered

40. What is the primary functional purpose of a header file (

.h

) in C programming?

1. To contain the main executable machine code
2. To securely hold function declarations, macros, and type definitions to be shared across source files
3. To define the aesthetic style of the console output
4. To physically link the compiled program to the host operating system

***** END OF PAPER *****